Lecture 8

In Class Exercises

Exercise 1

- What are the minimum and maximum number of elements in a heap of height h?

Exercise 1 (solution)

- Since a heap is an almost complete binary tree (complete at all levels except possibly the lowest), it has at most $2^{h+1} – 1$ elements (if it is complete) and at least 2^h elements (if the lowest level has just 1 element and the other levels are complete)

Exercise 2

- Show that an n-element heap has height $lg\ n$.

Exercise 2 (solution)

- Given an n-element heap of height h, we have:

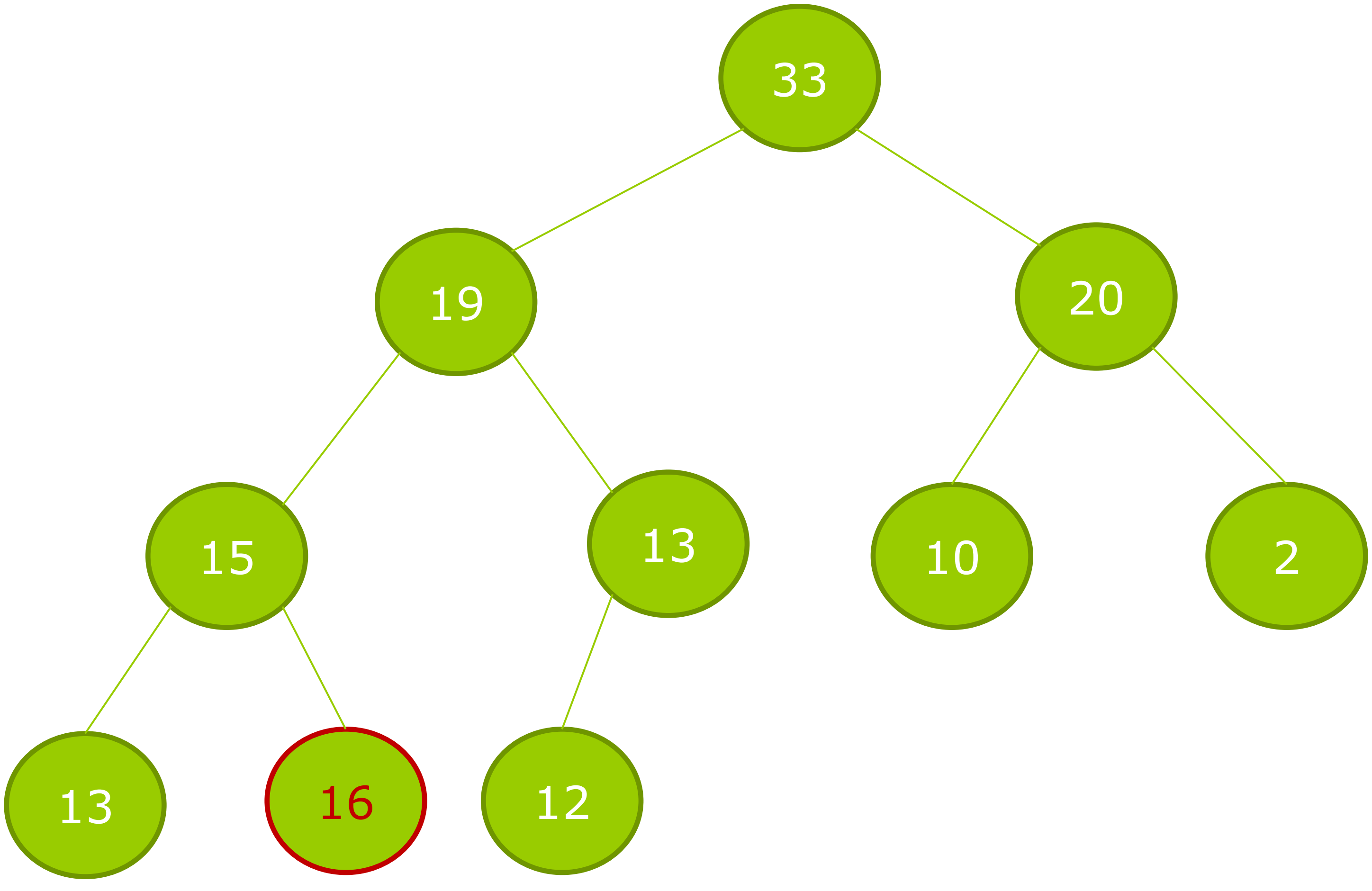
- $2^h \leq n \leq 2^{h+1} - 1 < 2^{h+1}$

- Therefore, $h \leq \log n \leq h+1$. Since h is an integer, $h = \lfloor \lg n \rfloor$

Exercise 3

- Build a heap from the following array and argue if it is a max heap:
- {33, 19, 20, 15, 13, 10, 2, 13, 16, 12}

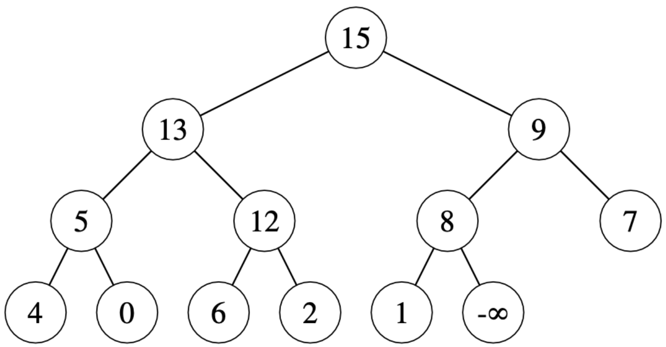
Exercise 3 (solution)



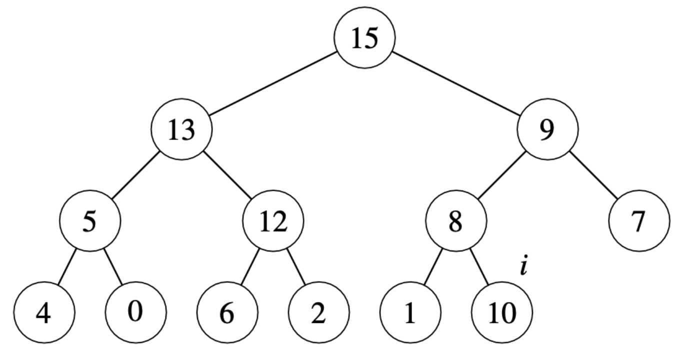
Exercise 4

- Suppose that the objects in a max-priority queue are just keys. Illustrate the operation of MAX-HEAP-INSERT(A,10) on the heap A = {15,13,9,5,12,8,7,4,0,6,2,1}.

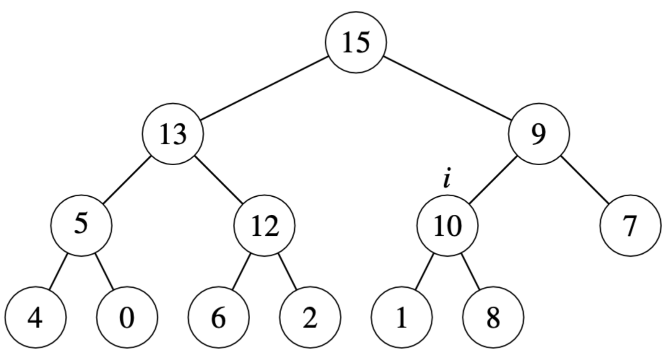
Exercise 4 (solution)



(a)



(b)



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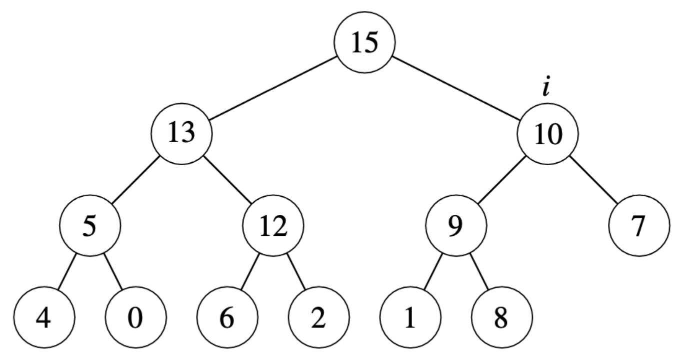

(d)